Science

Nature Walks & Scouting
General Science
Labs
Natural History
Nature Notebook

Science: Grade 4

GRADE(S): 4

About the Course

In level 4, learners begin to notice science in their community and begin to develop their own interests as they discover a variety of topics from the microscopic world, physical science, and more. A mixture of more and less challenging books is included in this course.

This course includes the following topic(s): General Science: Grade 4, Natural History: Grade 4, Labs: Grade 4, Nature Notebook: Grade 4, Nature Walks & Scouting: Grades 1-8

About Nature Walks & Scouting: Grades 1-8
Students spend time once each week on a longer excursion. Prompts can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

About General Science: Grade 4
Learners engage with microscopes, airplanes, and sound as they practice beginning lab skills and learn to construct bar graphs from their data. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

About Labs: Grade 4
This is the required lab component for level 4 General Science. This course includes our custom laboratory book which will build foundational and progressive skills and habits.

About Natural History: Grade 4
Learners encounter microscopic creatures, pursue the mysterious octopus, and watch the unfolding of life through primary succession. Coordinating afternoon activities are provided in Outdoor Work.

About Nature Notebook: Grade 4
Students spend time at least once each week recording their observations. Prompts that coordinate with curricular content can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

Science: Grade 4
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more “friends” in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge “by the way” and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right."

Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

Science: Grade 4
For fourth-grade students or possibly hungry third-graders or fifth-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, learners may stay at their appropriate level for General Science while combining Natural History and Outdoor Work, or teachers can adapt the reading level and laboratory content to meet their needs.

Nature Walks & Scouting: Grades 1-8
Learners may be combined and share prompts or follow the plan of their local scouting troop or natural history club. Support accessibility and sensory development as appropriate.

General Science: Grade 4
For fourth-grade students or possibly hungry third-graders or fifth-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.

Labs: Grade 4
The ideas and skills in this component are progressive, like math or grammar. Teachers should read the lab book thoroughly to understand what concepts might need to be supplemented should they choose a different sequence or a substitution.

Natural History: Grade 4
For approximately fourth-grade students based on a balance of complexity and reading level. However, learners may be freely combined or sequence reordered based on needs and preferences.

Nature Notebook: Grade 4
Learners may be combined and share prompts or follow their own interests. Support through varied media, scribing observations, or notebooking in tandem.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
1-8	Nature Walks & Scouting: Grades 1-8 1 time/week 30 min+	
4	General Science: Grade 4 1 time/week 20 min	The Elephant Scientist To Fly: The Story of the Wright Brothers The Universe in You: A Microscopic Journey
4	Labs: Grade 4 1 time/week 30 min	Science: Grade 4 Lab Book
4	Natural History: Grade 4 1 time/week 15 min	A World in a Drop of Water: Exploring with a Microscope Life on Surtsey: Iceland's Upstart Island The Octopus Scientists: Exploring the Mind of a Mollusk
4	Nature Notebook: Grade 4 1+ time/week 15 min+	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 4				
Natural History: Grade 4		General Science: Grade 4		Labs: Grade 4 Nature Notebook: Grade

		Nature Walks & Scouting: Grades 1-8		4

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 4
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Labs: Grade 4
☐ Preview science labs and purchase materials, or have students gather them. Print or shortcut any links, as desired.

Nature Notebook: Grade 4
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 4
Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

General Science: Grade 4
☐ Term 1: a museum with a microscopy exhibit
☐ Term 2: an airport observation deck and/or a museum with airplanes
☐ Term 3: a museum with a sound exhibit and a zoo

Natural History: Grade 4
☐ Term 1: a pond
☐ Term 2: an aquarium
☐ Term 3: a controlled burn site

Term Prep & Teacher Tips

Science: Grade 4
Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term

and/or any that support special topics chosen by students:

General Science: Grade 4
☐ Term 1: introduction to rocks p.743-745
☐ Term 2: one or more birds from p.62-142
☐ Term 3: sedimentary, igneous, and metamorphic rocks p.745-748

☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
• lighter or a match for candle
• penny
• flashlight
• a few salt crystals
• water
• table or counter with overhang
• hair dryer
• a few sheets of multipurpose paper
• long hallway for flying paper airplanes
• large-medium-sized bowl
• paper towels
• empty can or cup
• voice recorder, such as found on most electronic devices
• friendly animal, such as yours or a neighbor's pet (Term 3)
• printables
• optional: computer
• optional: sewing needle
• optional: metal coathanger
• optional: metal spoon
• optional: 2 pieces of string ~30cm

Natural History: Grade 4
☐ Term 1: water lily p.495-497, pondweed p.498-500
☐ Term 2: snails/mollusks p.416-421
☐ Term 3: one or more insects from p.301-367

Reminders

General Science: Grade 4
☐ Note that there is an activity during Lesson 1. Be sure to have your materials ready!

Natural History: Grade 4
☐ Students will need to collect a sample of pond water for use during Term 1: weeks 1, 8, and 10. There is a reminder to visit a pond in the Nature Notebook section of their Outdoor Work resource, but teachers may want to make a note in their calendars, as well.

Books & Resources

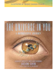
For book rationales and purchase options, click the Book List link or scan the QR code below.
∞ View Book List Details

Science: Grade 4

General Science: Grade 4

 The Elephant Scientist

 To Fly: The Story of the Wright Brothers

 The Universe in You: A Microscopic Journey

Labs: Grade 4

Science: Grade 4 Lab Book

Natural History: Grade 4

A World in a Drop of Water: Exploring with a Microscope

Life on Surtsey: Iceland's Upstart Island

The Octopus Scientists: Exploring the Mind of a Mollusk

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies

∞ View Supply List Details

Science: Grade 4

Alveary Grade 4 Science Kit

48 Prepared Microscope Slides for Beginner

Student Microscope

Nature Walks & Scouting: Grades 1-8

Wild Bird Seed

Bird Feeder

Hand Lens

Small Collection Containers

Small Basin

Dip Net

Student Microscope

General Science: Grade 4

Hand Lens

Household Items - General Science: Grade 4

Labs: Grade 4

Hand Lens

Small Collection Containers

Slides and Coverslips

Safety Goggles

Natural History: Grade 4

Slides and Coverslips

Quick Links

Click THIS text or scan the QR code for links.

Science: Grade 4

- Outdoor Work
- Microscope Coloring Book
- Seek app from iNaturalist
- SkyView Lite for iOS
- SkyView Lite for Android

Labs: Grade 4

- Grade 4 Lab Book
- Lab Notebook Examples

Science: Grade 4

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

• Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

• These can be viewed as alternative ways to engage.

Science: Grade 4

How To Teach

Introduce

Regardless of how many days are required to complete a particular activity, every science lab has the same flow, which follows the scientific method and is guided by the lab book.

• On day 1, learners read an introduction in their lab book. How does this lab activity relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is analogous to the conversation we might have when we begin something new in any subject, but they may need to dialogue as they learn to extend these skills to the laboratory.

• Once they have had a chance to think, they will compose a prelab narration to put these introductory ideas into their notebooks. The prompt in the lab is generalized and consistent, so they learn the habit. Eventually, they will learn to formulate this as a hypothesis. For example, a learner preparing for a lab about the use of insect repellent might write:

"I have read about some diseases that are spread by insects, like Lyme disease. I also know that my sister is allergic to some insect bites. Insect repellent contains pesticides to keep insects away. Some scientists worry about how pesticides affect wildlife. I am going to compare some different insect repellants in this lab to see if they really work."

• Written narration and composition are skills that they will build over time. These prelab narrations may seem short and even incomplete at first, but that is okay. If learners have difficulty or are easily frustrated, then provide them with support. Teachers may act as scribes or allow students to keep a digital notebook to type or use assistive technology, as appropriate.

• After they complete their prelab narration, the listed materials are collected. This gives the learner some responsibility to let the teacher know if something is missing or to remind the teacher if something needs to be purchased at the store.

• If these activities on day 1 do not fill the scheduled time, that is fine. They might use the additional time to familiarize themselves with the procedure, draw a picture from their book, or catch up on any other work. Some labs may instruct the student to begin on the same day.

Lab Procedure

• Then, students begin and follow the procedure (whenever prompted by the lab book). Note that the lesson plans guide teachers as to which sections are completed each week, and the lab book instructs learners when to take a break.

• The lab gives instructions for using their notebooks to create tables and figures, as needed. Do not allow this to become an obstacle. Do it with them, first having them watch and then having them copy or help when ready.

• If teachers choose to have learners record directly in the lab book or on a photocopy, then cut out and tape these into their lab books, so that they can see how the record is built.

• Learners may feel unsatisfied with their results. This is often part of the process. Come alongside, help them learn to be okay with uncertainty and questions, and teach them what to do with it in the next step.

Analysis & Conclusions

• The last step in the lab is to analyze the data and observations and draw conclusions from them. For some simpler labs, learners will complete their analysis and concluding (or postlab) narration on the last day of the lab procedure. For labs that are more involved, a separate day has been built in to allow adequate time for this.

• Similar to the prelab narration, the concluding narration is a chance to think about and put into words (or questions). This time, they are considering what they learned from the lab, what they could learn more about if they were to continue, and possibly how they would pursue that learning. Again, they might need support in the form of dialogue, a scribe, etc. For example, the above learner might write:

"It was clear that the Off and black pepper essential oil worked against ants because they would not even touch the line of repellent, but I wasn't sure about the Skin So Soft because it ran all over the place. I could test this one again with different insects or on a different surface."

• Depending on the interest of the learner and the priorities of the teacher, the student might be encouraged to spend more time on those ideas of what more they could learn, or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned.

• Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.

Science: Grade 4

[Click THIS text or scan the QR code for links.](#)

Term 1

WEEK 1

⬚ 15m Natural History: Grade 4 - Lesson 1

A New World

▯ Materials: A World in a Drop of Water

PREP: Read Teacher Tip

→ INTRO
Look at the picture on the cover. Have you ever seen such creatures before? In our book, the author will take us on a walk to find such creatures.

→ READ, NARRATE, & DISCUSS
p.3-5 "Let's go out" - "we have caught."

★ TEACHER TIP
Leave the microscope accessible throughout the term, so that samples can be observed easily in just a few minutes.

★ TEACHER NOTE
Students need to collect samples from a pond this week, as suggested in their nature notebook prompt. These will be used for several weeks DURING lesson activities.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1

⬚ 20m General Science: Grade 4 - Lesson 1

Zooming In

▯ Materials: The Universe in You, penny/dime, ruler, hand lens, image link

→ INTRO
What's the smallest Thing you've ever seen? Do you think there is anything smaller? Look at the picture on the cover. What do you think you are seeing? Did you know that there are such Things that are so small that we can't usually see them? This book is about the smallest Things we know of. You can use your ruler to track how small Things are for the first few pages.

→ READ
p.1-8 "The Calliope Hummingbird" - "of a vellus hair."

→ ACTIVITY
Can you see your skin cells or bacteria? How can we know they exist? Sometimes we can sense the effect of something we can't see, like the wind; sometimes we can use a tool to make it visible. To look closer at small Things, scientists use a tool called a microscope. We do 2 things with a microscope: magnify and focus.
• Look at ∞ Image Link: Penny & Salt or Dime & Salt
• This image has been magnified. Compare it to your coin. What does it mean to be magnified? How much bigger is it? How many of your coins could fit across this image?
• Did you notice that there are some salt crystals on the magnified coin? How much has the salt been magnified? (same as the coin)
• Use your hand lens to look at your coin. What details do you notice that you couldn't see without the hand lens? Look at the other side:
Penny: Can you find the little man between the columns of the building? Notice what happens as you move the penny and lens closer together/farther apart. What does it mean to focus?
Dime: Notice that the plants on either side of the torch are different. One is an olive and the other an oak. Can you tell the difference? Notice what happens as you move the dime and lens closer together/farther apart. What does it mean to focus?

→ NARRATE & DISCUSS

★ TEACHER TIP
Spend as much time as is wanted looking at the artwork in this book before going on to each successive page. Share the captioned facts, as desired.

★ TEACHER NOTE
Note that next week is an activity day. Check your supplies!

WEEK 1

⬚ 30m Labs: Grade 4 - Lesson 1

Lab: Meeting the Microscope

▯ Materials: Grade 4 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of Meeting the Microscope, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in the lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary. Spend remaining time gathering materials for next week.

★ TEACHER TIP
Pacing is only a suggestion. Students should engage with the lab at a pace that is appropriate for their abilities and interests.

WEEK 2

⬚ 15m Natural History: Grade 4 - Lesson 2

A Drop of Water Activity

▯ Materials: microscope, water sample, dropper, slide, cover slip

→ INTRO
Examine your pond water today.

→ ACTIVITY
• Make a few slides by transferring a drop of water to a slide and adding a cover slip. It helps to rest the edge of the coverslip on the slide, holding it at an angle. Then allow it to fall onto the drop of water.
• Observe your slide under low power.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Term 1

• If you find something under low power, try moving up to a higher power objective to see more. Adjust the focus, as needed.

→ NARRATE & DISCUSS
Use a nature journal to record what the student noticed. When you are finished, let the samples sit away from direct sunlight until you look at them again.
• Living things need food, water, and habitat. Would a living thing in the pond water have all of those?

WEEK 2

⬚ 20m General Science: Grade 4 - Lesson 2

The Microscope

Materials: video/image links, microscope, flashlight, slide, salt crystals

→ INTRO
What did we learn about in the last lesson? How do we see very small Things? What two things does a microscope do? Let's look at a very old microscope, and then we'll compare that to ours. Watch for him to show you the tube where the magnifying lenses are and the mirrors that direct the light.

→ VIEW
∞ Video Link: A 1750s Compound Microscope (5:06)

→ ACTIVITY
• Where is the tube and where are the lenses on your microscope? Refer to the diagram that came with your microscope. The one at the link may help for comparison. How much does the lens at your eyepiece (the ocular lens) magnify? How much do the lenses at the bottom of your tube (the objective lenses) magnify?
∞ Image Link: Compound Microscope Parts
• How did he use the mirror? Where is your light source? Place a slide with a few salt crystals on the stage. Does this direct light through the specimen or onto the surface? Look at the crystals and describe/draw what you see.
• Turn the light source off and instead shine the flashlight down onto the surface of the salt crystals. Adjust the flashlight to point as much light onto the surface as you can. Look at the crystals and describe how this is different from the lower light source.

→ NARRATE & DISCUSS

WEEK 2

⬚ 30m Labs: Grade 4 - Lesson 2

Lab: Meeting the Microscope

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Meeting the Microscope, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today, drawing, diagramming, or dictating in their lab notebook what they observe, just as they would in their nature journals.

WEEK 3

⬚ 15m Natural History: Grade 4 - Lesson 3

Inventing the Microscope

Materials: A World in a Drop of Water

→ INTRO
Recall what we read about in the last passage. Let's find out how the microscope was invented before we go further with it.

→ READ, NARRATE, & DISCUSS
p.6-9 "We are about" - "hundred years ago."
• Why do you think the scientists didn't believe Leeuwenhoek at first? What are some things that can be hard to believe because we can't see them?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

⬚ 20m General Science: Grade 4 - Lesson 3

Zooming in Further

Materials: The Universe in You

PREP: Read Teacher Tip

→ INTRO
What did the girl in this book see last time? Look at the pictures, as needed.

→ READ, NARRATE, & DISCUSS
p.9-end "The smallest hairs" - "the universe within."

★ TEACHER TIP
Students don't need to remember all of the scientific language. Just let them hear it "by the way."

Science: Grade 4

Click THIS text or scan the QR code for links.

Term 1

WEEK 3

⬚ **30m Labs: Grade 4 - Lesson 3**
Lab: Meeting the Microscope

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 3 of Meeting the Microscope, as directed in the Lab Book.

Suggested day 3: If students have a few last observations to make before completing their notebook, give them a few minutes to do so. Complete the postlab narration, as directed in the Analysis and Conclusion section of the lab. Teachers may scribe, journal in tandem, provide digital tools, or provide any other support as needed. If time remains, students may add a diagram of their microscope.

WEEK 4

⬚ **15m Natural History: Grade 4 - Lesson 4**
Amoeba

Materials: A World in a Drop of Water, microscope, water sample, dropper, slide, cover slip

PREP: Read Teacher Tip

→ INTRO
Recall Leeuwenhoek's discovery. Today we'll read about one of his animalcules, the amoeba.

→ READ, NARRATE, & DISCUSS
p.10-16 "To see what" - "to life again."

→ ACTIVITY
• Place a drop from your sample of the pond bottom onto a slide and add a coverslip.
• Look at it under the microscope. Is anything alive in there? Did you catch any amoeba?

→ SUPPLEMENTAL
∞ Video Link: Amoeba (0:47)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson, but they can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4

⬚ **20m General Science: Grade 4 - Lesson 4**
Imagining

Materials: poem link, image links

PREP: Read Teacher Tip

→ INTRO
We've been learning about very small Things. Tell me about Things too small to see. How can we know they are there? Some of the Things in your book (and in the lab) are so small that you won't be able to see them even with your microscope. Sometimes, when we first begin looking very closely and using a microscope, it can be hard to know what we are looking at. Read our poem for today. What do you think the author would say about the strange images we see under the microscope?

→ READ, DISCUSS, & REREAD
∞ Website Link: The Microbe

→ ACTIVITY
Even while we keep looking and asking questions, imagining what the image reminds us of can help us think about what we're seeing. Check out the imaginings at the link, and then draw or describe some of your own using the images in the Science Photo Gallery or some that you have seen under your microscope.
∞ Image Link: Microscope Imaginings
∞ Image Link: Science Photo Gallery

★ TEACHER TIP
Students who find copywork/dictation challenging may be anxious or unsure about today's activity. Encourage them to dictate, work in tandem, or even try a digital drawing program, if this is available.

• DEFINITIONS
sanguine: marked by eager hopefulness : confidently optimistic

WEEK 4

⬚ **30m Labs: Grade 4 - Lesson 4**
Lab: What's in Your Water?

Materials: Grade 4 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of What's in Your Water?, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Spend the remaining time gathering materials for next week.

★ TEACHER TIP
If needed, be sure to pour your agar plates for next week according to the package instructions.
∞ **Video Link:** Pouring Agar Plates (2:51)

Science: Grade 4

Click THIS text or scan the QR code for links.

Term 1

WEEK 5

⬚ 15m Natural History: Grade 4 - Lesson 5

Paramecium

Materials: A World in a Drop of Water, microscope, water sample, dropper, slide, cover slip

→ INTRO
Tell me what you remember about the amoeba. Today we'll read about another creature, the paramecium.

→ READ, NARRATE, & DISCUSS
p.17-24 "The amoeba" - "grow older."

→ ACTIVITY
• Make a slide from your sample of the pond's surface today. Is anything alive in there? Did you catch any paramecia?

→ SUPPLEMENTAL
It is not necessary to watch the whole video.
∞ Video Link: Paramecium (5:36)

★ TEACHER NOTE
The video link mentions adaptation and natural selection at 0:56 and 1:51, respectively. Teachers might choose to introduce the terms by the way, mute the narration, or skip this one.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5

⬚ 20m General Science: Grade 4 - Lesson 5

Louis Pasteur

Materials: video link

PREP: Read Teacher Tip

→ INTRO
Tell me what we were thinking about in the last lesson. The invention of the microscope allowed many scientists to begin looking closer and asking questions. Louis Pasteur is one of the most famous. Let's hear his story today.

→ VIEW, NARRATE, & DISCUSS
∞ Video Link: Louis Pasteur (10:36)

• Pasteur said, "Do not let yourself be tainted with barren skepticism." What does that mean? How do we ask questions, like the poet who wrote the Microbe suggests, without being "tainted with barren skepticism"?

★ TEACHER TIP
Are your learner(s) demonstrating more ease in using the microscope? More interest/ease in describing what they see with it or how it works? If not, consider practicing together more often, exploring extra helpings, or taking a field trip to a museum or lab with microscopes. Reach out through Contact Us if you need help.

WEEK 5

⬚ 30m Labs: Grade 4 - Lesson 5

Lab: What's in Your Water?

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of What's in Your Water?, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today, drawing, diagramming, or dictating in their lab notebook what they observe, just as they would in their nature journals.

WEEK 6

⬚ 15m Natural History: Grade 4 - Lesson 6

Algae

Materials: A World in a Drop of Water, microscope, water sample, dropper, slide, cover slip

PREP: Read Teacher Tip

→ INTRO
Recall what you learned in the last passage. Today we will read about algae, which belong to a very diverse group of living things. Scientists still disagree about how to think about them. In fact, the author refers to algae as plants, but today they are called protists.

→ READ, NARRATE, & DISCUSS
p.25-30 "Remember the thin" - "on their own."

→ ACTIVITY
• Look at your water sample from the surface of the pond. There is probably some algae growing on the top.
• Make a slide from this sample, being sure to capture a bit of this algae. What do you notice?

★ TEACHER TIP
Are your learner(s) demonstrating more ease in using the microscope? More interest/ease in describing the tiny creatures they see with it? If not, consider practicing together more often, exploring extra helpings, or taking a field trip to a museum or lab with microscopes.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6

⬚ 20m General Science: Grade 4 - Lesson 6

Microscopy Questions Today

Materials: video link

→ INTRO
What have we been learning about? Who was Louis Pasteur, and what kinds of questions did he ask? What kinds of questions do you think people ask with their microscopes today? For the next

★ TEACHER NOTE
The lesson video mentions evolution and adaptation during 5:37-6:40. Teachers might choose to introduce the terms by the way, forward through that portion, or watch without some or all of the narration.

Term 1

few weeks, we'll learn about some of today's questions.

→ VIEW, NARRATE, & DISCUSS
∞ Video Link: We Don't Know Why Moth Wings Glow (10:09)

WEEK 6

⬚ 30m Labs: Grade 4 - Lesson 6
Lab: What's in Your Water?

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 3 of What's in Your Water?, as directed in the Lab Book.

Suggested day 3: Students complete the Analysis and Conclusion instruction in the lab book, recording in their lab notebook just as they would in their nature journals. Teachers may scribe, journal in tandem, provide digital tools, or provide any other support as needed.

WEEK 7

⬚ 15m Natural History: Grade 4 - Lesson 7
Euglena

Materials: A World in a Drop of Water

→ INTRO
Recall what you learned in the last passage. Today we will read about another creature, called euglena.

→ READ, NARRATE, & DISCUSS
p.31-33 "Swimming quickly" - "called protists."

★ TEACHER NOTE
Students need to collect samples from a pond again next week, as suggested in their nature notebook prompt.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

⬚ 20m General Science: Grade 4 - Lesson 7
Microscopy Questions Today

Materials: video link

→ INTRO
What did we see last week? Today we have another question that scientists are using microscopes to investigate.

→ VIEW, NARRATE, & DISCUSS
∞ Video Link: The Cryptic Origins of Yogurt (10:21)

WEEK 7

⬚ 30m Labs: Grade 4 - Lesson 7
Lab: What's in Your Mouth?

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 1 of What's in Your Mouth?, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Spend the remaining time gathering materials for next week.

WEEK 8

⬚ 15m Natural History: Grade 4 - Lesson 8
Rotifer

Materials: A World in a Drop of Water

PREP: Read Teacher Tip

→ INTRO
Recall what you learned in the last passage. Look at the picture on p.35. The title says this is an animal with wheels. Can you see where the wheels are?

→ READ, NARRATE, & DISCUSS
p.34-37 "All the animals" - "back to life."

• Rotifers are the same size as many of the protists that we met this term, yet they are animals. Why?

→ SUPPLEMENTAL
∞ Video Link: Rotifers Under the Microscope (3:50)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Science: Grade 4

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 8

⬚ 20m General Science: Grade 4 - Lesson 8

Microscopy Questions Today

Materials: video link

→ INTRO
Tell me about our video from last week. Let's learn about one more question today.

→ VIEW, NARRATE, & DISCUSS
∞ Video Link: Tiny Mysteries from the Black Sea (8:07)

WEEK 8

⬚ 30m Labs: Grade 4 - Lesson 8

Lab: What's in Your Mouth?

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of What's in Your Mouth?, as directed in the Lab Book.
∞ Video Link: How to Dry Mount Cheek Cells (4:07)

Suggested day 2: Students conduct the Procedure today. They will need help fixing and staining their slides. Then they may draw, diagram, or dictate in their lab notebook what they observe, just as they would in their nature journals. Today they will continue to the end, completing the short Analysis and Conclusion with a brief narration.

WEEK 9

⬚ 15m Natural History: Grade 4 - Lesson 9

Hydra

Materials: A World in a Drop of Water, image link, microscope, water sample, dropper, slide, cover slip, forceps

→ INTRO
Remind me what we read in the last passage. Look at the picture on p.39. Now look at the image of the Greek artifact at the link. Can you see why this animal was named after the monster?
∞ Image Link: Hydra of Greek Mythology

→ READ, NARRATE, & DISCUSS
p.38-44 "Many of" - "into new hydras."

→ ACTIVITY
• Use the forceps to pick up a small piece of your aquatic plant. Examine it to see if you can see anything with the naked eye.
• If it is a thin piece, then place it onto a glass slide and add a cover slip. If it is too thick for the microscope, then use the medicine dropper to rinse the underside off onto the slide with a drop or two of pond water. Then add the cover slip.
• Look closely at your sample under the microscope. What do you see? Is a rotifer there? What about a hydra? Are any of the protists that we previously met there?

→ SUPPLEMENTAL
∞ Video Link: Hydra Under the Microscope (4:25)

★ TEACHER NOTE
The reproduction of hydra is described on p.44: "Hydras can... new hydras." Teachers might allow it to pass by the way, decide if they would like to discuss, or omit the last paragraph.

★ TEACHER NOTE
Students need to collect samples from a pond again next week, as suggested in their nature notebook prompt.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9

⬚ 20m General Science: Grade 4 - Lesson 9

The Universe in You

Materials: The Universe in You

→ INTRO
Tell me about some of the questions that scientists are using microscopes to investigate. Now that you've seen your own cells in the lab, let's read our book one more time.

→ READ, NARRATE, & DISCUSS
The Universe in You (whole book)

WEEK 9

⬚ 30m Labs: Grade 4 - Lesson 9

Lab: Explore and Discover!

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 1 of Explore and Discover!, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Students may collect their materials and begin the Procedure, as time permits.

★ TEACHER TIP
If needed, be sure to pour your agar plates for next week according to the package instructions.
∞ **Video Link:** Pouring Agar Plates (2:51)

Science: Grade 4

Click THIS text or scan the QR code for links.

Term 1

WEEK 10

⬚ 15m Natural History: Grade 4 - Lesson 10

Flatworm

Materials: A World in a Drop of Water

PREP: Read Teacher Tip

→ INTRO
Remind me what we read in the last passage. Our last tiny creature from the pond is the flatworm.

→ READ, NARRATE, & DISCUSS
p.45-51 "Although we've" - "smaller and smaller."

→ SUPPLEMENTAL
∞ Image Link: Missouri Department of Conservation: Planaria

★ TEACHER NOTE
Planarian sexuality is described on p.48: "Planarians have... way of reproducing." Teachers might allow it to pass by the way, decide if they would like to discuss, or omit this paragraph.

★ TEACHER TIP
The image link will give you an idea of what planaria look like on the underside of rocks. They can be quite small, and their color can be a range of grays and browns.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10

⬚ 20m General Science: Grade 4 - Lesson 10

Zooming In Even More

Materials: website link, printable

PREP: Read Teacher Tip

→ INTRO
What are some of the most interesting Things you've looked at with your microscope? What else would you like to see? You may have noticed that when you zoom in as much as you can, the image starts to get blurry. Once they zoom in that far, scientists have to use another kind of microscope to zoom in even more. The Science Photo link contains a gallery of images using these special microscopes. The scientist who prepared these images added color to help draw our attention to certain features. Explore the gallery.

→ VIEW
∞ Website Link: Science Photo Gallery

→ ACTIVITY
The links below contain a collection of original images without color that a scientist shared especially for you. You may choose images from this collection to add your own color. Which features do you find most interesting? Which ones do you want your viewer to notice?
∞ Printable Link: Electron Microscope Coloring Pages

★ TEACHER TIP
Students who find copywork/dictation challenging may be anxious or unsure about today's coloring activity. Encourage them to dictate, work in tandem, or even try a digital coloring program, if this is available.

WEEK 10

⬚ 30m Labs: Grade 4 - Lesson 10

Lab: Explore and Discover!

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Explore and Discover!, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today, drawing, diagramming, or dictating in their lab notebook what they observe, just as they would in their nature journals.

WEEK 11

⬚ 15m Natural History: Grade 4 - Lesson 11

Last Look (For Now)

Materials: A World in a Drop of Water, microscope, water sample, dropper, slide, cover slip

PREP: Read Teacher Tip

→ INTRO
Remind me what we read in the last passage. Today we'll wrap up and then revisit any samples that you want to take one last look at.

→ READ, NARRATE, & DISCUSS
p.52-56 "While we are" - "strange unseen world."

• Use the microscope to revisit any of your samples before you clean out the containers!

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether the microscope or any of the new creatures they have met are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11

⬚ 20m General Science: Grade 4 - Lesson 11

Catch Up

PREP: Read Teacher Tip

→ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly, whether the microscope, people, or Things they have seen are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on

Term 1

		time, more reading support, less business, etc? Reach out through Contact Us if you need help.
WEEK 11	⬚ **30m Labs: Grade 4 - Lesson 11** *Lab: Explore and Discover!* Materials: Grade 4 Lab Book and materials listed within **➔ LAB DAY** Complete day 3 of Explore and Discover!, or catch up on any other lab. Suggested day 3: Students complete the Analysis and Conclusion instruction in the lab book, finishing any remaining observations and completing the postlab narration. Teachers provide whatever support is needed.	
WEEK 12	⬚ **15m Natural History: Grade 4 - Lesson 12** *➔ EXAMS* Answer question(s) related to course. Questions will come from: A World in a Drop of Water	• **OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	⬚ **20m General Science: Grade 4 - Lesson 12** *Exam Week* **➔ EXAMS** Answer question(s) related to course. Questions will come from: The Universe in You and multimedia content	
WEEK 12	⬚ **30m Labs: Grade 4 - Lesson 12** *Lab Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Grade 4 Lab Book	

Science: Grade 4

[Click THIS text or scan the QR code for links.]

Term 2

WEEK 1

⬚ 15m Natural History: Grade 4 - Lesson 13

An Adventure to the Pacific

Materials: The Octopus Scientists

PREP: Read Teacher Tip

→ INTRO
Look at the picture on the cover. What do you know about octopuses? Octopuses live all over the world, but this book is going to tell us about the search for them on the island of Moorea in the Pacific Ocean. Look at the map on the page following the Table of Contents, noticing the island's location on the lower left. Can you find the location on your globe?

→ READ, NARRATE, & DISCUSS
p.1-4 “The ocean is” - “eaten themselves.”

• What kinds of questions would you ask if you were an octopus scientist?

→ SUPPLEMENTAL
∞ Image Link: Aquarium of the Pacific: Day Octopus
∞ Video Link: YouTube: Natural History Museum (1:05)

→ NOTE
We are not going to read the next few paragraphs during lessons ("What good" - "about marine animals."), but if students are interested, feel free to explore this extra passage.

★ TEACHER TIP
This author writes a rich and interesting narrative that weaves together observations about the creature, their ecosystem, the scientists who study them, and the process of science itself. This richness can sometimes feel complex when narrating. If so, teach students to first notice what the purpose of the passage is. Today readers are introduced to the location and the creature that is so difficult to find. How did she begin telling us about that? And then?

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1

⬚ 20m General Science: Grade 4 - Lesson 13

Inspiration

Materials: To Fly

PREP: Read Teacher Tip

→ INTRO
Look at the picture on the cover. Do you know who the Wright Brothers are? What do you know about them? Read and discuss the poem called "Crazy Boys" in the early pages of the book.

→ READ, NARRATE, & DISCUSS
p.7 "People had always" - "For a time."

• How do you think the toys that inspired the Wrights worked? Can you tell from the picture? Compare to this image from the Smithsonian Museum:
∞ Image Link: Smithsonian: Orville's Sketch of Penaud's Helicopter

→ SUPPLEMENTAL
Notice how many times the craftsman in the video fails before he gets it right, but how wonderfully it works when he does.
∞ Video Link: YouTube: Making a Penaud Helicopter (5:04)

★ TEACHER TIP
As you read, notice the purpose of the passage. Then, if students struggle to narrate, you can prompt them by helping them identify the purpose. Is it describing a scene or a person? Telling a sequence of events? Explaining an idea?

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 1

⬚ 30m Labs: Grade 4 - Lesson 13

Lab: Simple Machine Scavenger Hunt

Materials: Grade 4 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of Simple Machine Scavenger Hunt, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary. Any remaining time can be spent copying the drawings from the Lab Book into their notebook.

★ TEACHER TIP
Pacing is only a suggestion. Students should engage with the lab at a pace that is appropriate for their abilities and interests.

WEEK 2

⬚ 15m Natural History: Grade 4 - Lesson 14

Getting Our Feet Wet

Materials: The Octopus Scientists

PREP: Read Teacher Tip

→ INTRO

★ TEACHER TIP
If students need support with writing or diagramming, teachers should model, scribe, or offer whatever assistance is appropriate.

• OUTDOOR WORK
Weekly walks and daily outdoor

Term 2

Recall what you read in the last passage. Where has the author taken us, and what are we doing? Do you think they will find an octopus today?

→ READ, NARRATE, & DISCUSS
p.5-6 “Dripping with” - “Where is it now?”

• Build a food web for the octopus based on what you have read so far.

→ NOTE
Read "Meet the Octopus Team" p.7-10, or "An Octet of Octo Facts" p.11 for interest, if desired.

time are vital! Resources in Quick Links.

WEEK 2

⬚ 20m General Science: Grade 4 - Lesson 14
Observing Motion

Materials: To Fly

PREP: Read Teacher Tip

→ INTRO
Recall how the story began. Do you think the propeller on their toy helicopter is a simple machine? If so, which might it be?

→ READ, NARRATE, & DISCUSS
p.8 "Wilbur and Orville" - "than flat ones."

• Do you think the kites that Orville built were simple machines? If so, which one? If not, why not? Spend any remaining lesson time noticing objects or tools that make work and motion easier. This is a good indication that there is probably a simple machine at work.

★ TEACHER TIP
Hinges, knives, jar lids, any piece of playground equipment, the pulley on a flagpole, garage doors, stairs, scissors, etc., are all simple machines. Help students notice that they are everywhere while discussing.

WEEK 2

⬚ 30m Labs: Grade 4 - Lesson 14
Lab: Simple Machine Scavenger Hunt

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Simple Machine Scavenger Hunt, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today. They may need help learning how to use the grid lines in their notebook to create a table for data collection. Students not ready to do so may use the table in the Lab Book and then cut and paste into their notebook.

WEEK 3

⬚ 15m Natural History: Grade 4 - Lesson 15
On the Hunt

Materials: The Octopus Scientists

→ INTRO
Remind me what we read in the last passage. So, they are on the hunt. Let's see how they are going to find this creature.

→ READ, NARRATE, & DISCUSS
p.13-16 “What do we” - “anybody yet.”

• What would you say is their key strategy to finding the octopus? Is there a creature nearby that you would like to learn more about? If you wanted to look for it using the same strategy as our scientists, how would you go about your search?
• Describe some Thing in nature that you would like to know better.

→ SUPPLEMENTAL
∞ Image Link: National Geographic: Butterfly Fish
∞ Image Link: National Geographic: Pufferfish

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

⬚ 20m General Science: Grade 4 - Lesson 15
Learning Together

Materials: To Fly

→ INTRO
Tell me about the Wrights so far. Let's find out what happens when they grow up.

→ READ, NARRATE, & DISCUSS
p.10-13 "In time, Orville" - "the right answer."

• Disagreeing and arguing seemed to help Orville and Wilbur. Do you feel the same way when someone disagrees with you? If so, how does it help you? If not, why not? When you disagree with someone, do you feel like you are trying to help them?

Science: Grade 4

[Click THIS text or scan the QR code for links.]

Term 2

WEEK 3

⬚ 30m Labs: Grade 4 - Lesson 15
Lab: Simple Machine Scavenger Hunt

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 3 of Simple Machine Scavenger Hunt, as directed in the Lab Book.

Suggested day 3: Students finish the lab today using the Analysis and Conclusions section. They may need help learning how to construct a bar graph using the grid lines in their notebook, if they have not done this before. Instructions are provided in the Lab Book. If needed, students may use the table in the Lab Book and then cut and paste into their notebook. Then complete the postlab narration, as directed.

WEEK 4

⬚ 15m Natural History: Grade 4 - Lesson 16
Hoping for Serendipity

Materials: The Octopus Scientists

→ INTRO
Recall what you read in the last passage. Do you think they will find an octopus at this site?

→ READ, NARRATE, & DISCUSS
p.16-17 "The afternoon's" - "really good."

• Tell or draw about a time when you were hunting for some Thing in nature.
• Discuss how frustrating it can be when you don't find the creature you are looking for on a walk or when any activity doesn't turn out the way you had hoped. What helps you to persevere?

→ NOTE
Read "Criobe" p.18 for interest, if desired.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4

⬚ 20m General Science: Grade 4 - Lesson 16
Kites Become Gliders

Materials: To Fly

→ INTRO
Where did we leave the Wrights? Recall what happened. How do you think they got from bicycles to airplanes?

→ READ, NARRATE, & DISCUSS
p.15-18 "In 1896, Orville" - "was build it."

• Describe the 3 movements described in the passage. You can tell, draw, or use your body to do this. Show the special mechanism that would allow the Wrights to control the movements.

WEEK 4

⬚ 30m Labs: Grade 4 - Lesson 16
Lab: What's an Aerofoil

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 1 of What's an Aerofoil, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Any remaining time can be spent collecting the materials and copying the drawing from the Lab Book into their notebook.

WEEK 5

⬚ 15m Natural History: Grade 4 - Lesson 17
Continuing the Hunt

Materials: The Octopus Scientists

PREP: Read Teacher Tip

→ INTRO
Recall what you read in the last passage. Let's see how they decide to continue today.

→ READ, NARRATE, & DISCUSS
p.21-24 "Wet bathing suits" - "a dry towel."

→ NOTE
Read "How Smart is an Octopus?" p.25 for interest, if desired.

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity with this strange creature? Greater understanding that science is often frustrating? If not, consider discussing big ideas during everyday life, exploring extra helpings, or taking a field trip to an aquarium.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

[Click THIS text or scan the QR code for links.]

Term 2

WEEK 5

⬚ 20m General Science: Grade 4 - Lesson 17
Experiments

Materials: To Fly

PREP: Read Teacher Tip

→ INTRO
Recall what happened in the last passage.

→ READ, NARRATE, & DISCUSS
p.19-22 "Wilbur wrote to" - "problems of flight."

• There were many highs and lows during the Wrights' engineering process. Have you ever worked at something that got you both excited and frustrated? Did you want to give up?

★ TEACHER TIP
Are your learner(s) demonstrating more ease in exploring how Things work? More interest/ease in thinking about the Wright Brothers or their work? If not, consider playing with mechanical toys together, exploring extra helpings, or taking a field trip to a park to observe the equipment. Reach out through Contact Us if you need help.

WEEK 5

⬚ 30m Labs: Grade 4 - Lesson 17
Lab: What's an Aerofoil

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of What's an Aerofoil, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today, drawing, diagramming, or dictating in their lab notebook what they observe, just as they would in their nature journals.

WEEK 6

⬚ 15m Natural History: Grade 4 - Lesson 18
Adjusting

Materials: The Octopus Scientists

PREP: Read Teacher Tip

→ INTRO
Recall what you read in the last passage. What do you think will happen today?

→ READ & NARRATE
p.27 “We have been” - “in the water.”

→ READ, NARRATE, & DISCUSS
p.28-31 “The next day” - “hard to study.”

• Notice how frustrations and enjoyment sometimes go together. What is something that is frustrating to you? Think of an upside to that activity.

→ NOTE
Read "Making Friends with an Octopus" p.32 for interest, if desired.

★ TEACHER TIP
Sometimes more complex readings can be broken into smaller parts. In this passage, the first part describes how they plan to go about their study, while the second part tells the story of their first attempt to implement the plan. This is a great opportunity to help students notice that the way we narrate can 'adjust' to the passage!

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6

⬚ 20m General Science: Grade 4 - Lesson 18
Using the Right Information

Materials: To Fly

→ INTRO
Recall what happened in the last passage.

→ READ, NARRATE, & DISCUSS
p.24-27 "The brothers began" - "self-propelled flying machine."

• What do you think helped the Wrights to persevere and keep going even though they had been so frustrated in the last reading?

WEEK 6

⬚ 30m Labs: Grade 4 - Lesson 18
Lab: What's an Aerofoil

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 3 of What's an Aerofoil, as directed in the Lab Book.

Suggested day 3: Students finish the lab today, as guided in the Analysis and Conclusions section.

Science: Grade 4

[Click THIS text or scan the QR code for links.]

Term 2

WEEK 7

⬚ 15m Natural History: Grade 4 - Lesson 19

Grouping Creatures to Better Understand

Materials: The Octopus Scientists

PREP: Read Teacher Tip

→ INTRO
Recall what you read in the last passage.

→ READ, NARRATE, & DISCUSS
P35-38 “Preserved pufferfish.” - “the next day.”

• What plants or animals have you noticed that seem to belong to the same family? Even within that family, have you noticed that some seem to be more similar to each other? Try to think of examples, such as some birds eat seeds while others eat bugs or some cats purr while others roar.

★ TEACHER TIP
In this passage, Jennifer is teaching about the concept of classification within the context of the story. If students have difficulty narrating who said what and recalling the finer details, it may help to first notice that larger purpose. Then students can narrate or describe the concept and the process demonstrated by the scientists rather than trying to recall a sequence of events.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

⬚ 20m General Science: Grade 4 - Lesson 19

Designing a Flyer

Materials: To Fly

→ INTRO
Recall what happened in the last passage.

→ READ, NARRATE, & DISCUSS
p.28-32 "During the winter" - "a homemade engine?"

WEEK 7

⬚ 30m Labs: Grade 4 - Lesson 19

Lab: Paper Airplanes

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 1 of Paper Airplanes, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Any remaining time can be spent collecting the materials and copying the drawing from the Lab Book into their notebook.

WEEK 8

⬚ 15m Natural History: Grade 4 - Lesson 20

Glimpsing the Octopus

Materials: The Octopus Scientists

PREP: Read Teacher Tip

→ INTRO
Recall what you read in the last passage.

→ READ, NARRATE, & DISCUSS
p.38-41 “For most of” - “so were we!”
• Describe a day in the life of an octopus scientist.

→ SUPPLEMENTAL
∞ Video Link: YouTube: Octopus Camouflage (1:46)

→ NOTE
Read "How Octopuses Change Color" p.42-43 for interest, if desired.

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

⬚ 20m General Science: Grade 4 - Lesson 20

The First Attempt

Materials: To Fly

PREP: Read Teacher TIp

→ INTRO
Recall what happened in the last passage.

→ READ, NARRATE, & DISCUSS
p.33-34 "On Monday, December" - "a successful flight."

→ SUPPLEMENTAL
Examine the photographs in the link, especially photos 1, 2, and 5. Notice what the differences

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

Term 2

were between their gliders and the airplane. If interested, draw or diagram any or all of them, showing the important features.
∞ Image Link: Wright Brothers Memorial

WEEK 8

30m Labs: Grade 4 - Lesson 20
Lab: Paper Airplanes

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Paper Airplanes, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today. They may need help learning how to use the grid lines in their notebook to create a table for data collection. If needed, students may use the table in the Lab Book and then cut and paste into their notebook.

WEEK 9

15m Natural History: Grade 4 - Lesson 21
Where They Aren't

Materials: The Octopus Scientists

→ INTRO
Recall what you read in the last passage.

→ READ, NARRATE, & DISCUSS
p.45-49 “We celebrate” - “need to know.”
• Science can be a frustrating process! Notice the ways that these scientists overcome their frustration. Can you think of a time when you were able to problem-solve to overcome frustration?

→ SUPPLEMENTAL
∞ Image Link: Malta National Aquarium: Picasso Triggerfish

→ NOTE
Read Rainforests of the Ocean p.51-52 for interest, if desired.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9

20m General Science: Grade 4 - Lesson 21
The First Flight

Materials: To Fly

→ INTRO
Recall what happened in the last passage. Do you think they will try again?

→ READ, NARRATE, & DISCUSS
p.36-41 "It took two" - "Aviation had begun."

• Their friend said they seemed like two people "who weren't sure they'd ever see each other again." What does it say about their characters that they kept trying even though they knew they could die? Why would they do this?

→ SUPPLEMENTAL
∞ Image Link: Library of Congress: First Flight

• NATURE NOTEBOOK
Prompt for General Science: Grade 4 in Outdoor Work Quick Link.

WEEK 9

30m Labs: Grade 4 - Lesson 21
Lab: Paper Airplanes

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 3 of Paper Airplanes, as directed in the Lab Book.

Suggested day 3: Many students want to collect another set of flight data or compare it with another airplane design. If so, let them do so and test their ideas. Challenge them to think about what is causing any differences that they see in their data. Students who are ready to move on may skip ahead to next week.

WEEK 10

15m Natural History: Grade 4 - Lesson 22
Joy!

Materials: The Octopus Scientists

→ INTRO
Recall what you read in the last passage.

→ READ, NARRATE, & DISCUSS
p.53-57 “By the next” - “Joy!”
• How has their understanding of the geography of the island become important to their hunt?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Term 2

Have you ever noticed creatures that like a certain kind of land feature?

→ NOTE
Read "Octopus Influence" p.58 for interest, if desired.

WEEK 10

⬚ 20m General Science: Grade 4 - Lesson 22
Lessons from their Story

Materials: To Fly

→ INTRO
Recall what happened in the last passage. What do you think the world thought about this news?

→ READ, NARRATE, & DISCUSS
p.42-44 "The brothers packed" - "actually did it."

• Everyone has strengths and challenges, even the Wright Brothers. Talk about how you have noticed this in yourself or your friends and family.

• NATURE NOTEBOOK Prompt for General Science: Grade 4 in Outdoor Work Quick Link.

WEEK 10

⬚ 30m Labs: Grade 4 - Lesson 22
Lab: Paper Airplanes

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 4 of Paper Airplanes, as directed in the Lab Book.

Suggested day 4: Students finish the lab today using the Analysis and Conclusions section. They may need help learning how to construct a bar graph using the grid lines in their notebook. Instructions are provided in the Lab Book. If needed, students may use the table in the Lab Book and then cut and paste into their notebook. Then complete the postlab narration, as directed.

WEEK 11

⬚ 15m Natural History: Grade 4 - Lesson 23
A Lot to Learn

Materials: The Octopus Scientists

PREP: Read Teacher Tip

→ INTRO
Recall what you read in the last passage.

→ READ, NARRATE, & DISCUSS
p.61-65 “The next morning” - “a splendid hostess.”

→ NOTE
Depending on time and student interest, finish conclusion p.65-66 “David named” - “what it’s about!” OR just p.66 “In all” - “what it’s about!”

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether the octopus or even the practice of doing science itself has become more comfortable and friendly! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11

⬚ 20m General Science: Grade 4 - Lesson 23
Catch Up

PREP: Read Teacher Tip

→ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly, whether the airplane, the Wright Brothers, any mechanical Things, or any Things that they have met are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us if you need help.

• FIELD TRIP
Visit a location that has airplanes. The Wright Brothers National Memorial, an Air and Space museum, or an airport observation deck are examples.

WEEK 11

⬚ 30m Labs: Grade 4 - Lesson 23
Lab: Wings for Flight

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete the Wings for Flight activity, as directed in the Lab Book, or use this day to catch up on any other work.

Science: Grade 4

[Click THIS text or scan the QR code for links.]

Term 2

WEEK 12	⬚ **15m Natural History: Grade 4 - Lesson 24** *→ EXAMS* Answer question(s) related to course. Questions will come from: The Octopus Scientists	• **OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	⬚ **20m General Science: Grade 4 - Lesson 24** *Exam Week* **→ EXAMS** Answer question(s) related to course. Questions will come from: To Fly	
WEEK 12	⬚ **30m Labs: Grade 4 - Lesson 24** *Lab Exam Week* **→ EXAMS** Answer question(s) related to the course. Questions will come from: Grade 4 Lab Book	

Science: Grade 4

Click THIS text or scan the QR code for links.

Term 3

WEEK 1

⬚ 15m Natural History: Grade 4 - Lesson 25
Arrival

▯ Materials: Life on Surtsey

PREP: Read Teacher Tip

→ INTRO
Look at the picture on the cover. What do you notice about it? Locate Surtsey on the map on p.1. Surtsey is a very young island. This is a story about scientists who are studying its growth and development, but they only get to visit the island for a single week each year.

→ READ, NARRATE, & DISCUSS
p.1-5 "Drengur Ólafsson" - "for a while."

→ SUPPLEMENTAL
∞ Video Link: What is a Volcano? (3:14)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1

⬚ 20m General Science: Grade 4 - Lesson 25
It Reminds Me Of

▯ Materials: The Elephant Scientist, globe

→ INTRO
Look at the picture on the cover and the title. Have you seen elephants at a zoo? What do you remember about them? Do you think they communicate with each other? If so, how? This is a story about scientists who are trying to explain how they communicate and what they are saying. Most of the elephants in this book live at Etosha National Park and Mushara Waterhole. You can find it on the map on the page just before the Table of Contents. Can you find the location on your globe?

→ READ, NARRATE, & DISCUSS
p.1-3 "Caitlin O'Connell peered" - "sense their environment."

WEEK 1

⬚ 30m Labs: Grade 4 - Lesson 25
Lab: Seeing Sound

▯ Materials: Grade 4 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of Seeing Sound, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary. Any remaining time can be spent gathering materials and/or copying the drawings from the Lab Book into their notebook.

★ TEACHER TIP
Pacing is only a suggestion. Students should engage with the lab at a pace that is appropriate for their abilities and interests.

WEEK 2

⬚ 15m Natural History: Grade 4 - Lesson 26
The Volcano Erupts

▯ Materials: Life on Surtsey

→ INTRO
Recall what you read in the last passage.

→ READ, NARRATE, & DISCUSS
p.5-6 "From the moment" - "childhood home"
• Can you think of reasons that these scientists would be so determined to study Surtsey?

→ SUPPLEMENTAL
∞ Video Link: Submarine Volcanoes (3:12)

→ NOTE
Read "Sizing Up Surtsey" p.7 for interest, if desired.

★ TEACHER NOTE
The supplemental video mentions the age of the Earth at 2:54. Teachers might choose to allow it to pass by the way, consider how they want to discuss with their students, stop the video early, or skip this one.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

⬚ 20m General Science: Grade 4 - Lesson 26
Becoming a Scientist

▯ Materials: The Elephant Scientist

PREP: Read Teacher Tip

→ INTRO
Recall what happened in the last passage.

→ READ & NARRATE

★ TEACHER TIP
Some students may be able to read the entire passage and narrate the scientist's journey, but others may benefit from having it broken into smaller portions, as it is presented here. Do whatever is most appropriate for the student(s).

Term 3

p.4-6 "Caitlin O'Connell studies" - "her animal studies."

→ READ, NARRATE, & DISCUSS
p.7-10 "After working with" - "she never anticipated."

• Do you think it would be challenging to study and learn about something that you cannot see? Why or why not? How can technology, like you see in the link, help scientists study the invisible?
∞ Video Link: What does Sound look like? (2:32)

WEEK 2

30m Labs: Grade 4 - Lesson 26

Lab: Seeing Sound

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Seeing Sound, as directed in the Lab Book.

Suggested day 2: Students conduct the Procedure today. They may draw, diagram, or dictate in their lab notebook what they observe, just as they would in their nature journals. Then they will continue to the end, completing the short Analysis and Conclusion with a brief narration.

WEEK 3

15m Natural History: Grade 4 - Lesson 27

To Study Surtsey

Materials: Life on Surtsey

→ INTRO
Recall what you read in the last passage.

→ READ, NARRATE, & DISCUSS
p.10-13 "In 1969" - "Home.'"
• How do you think it would feel to be chosen to study an isolated, undeveloped island?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

20m General Science: Grade 4 - Lesson 27

To Be an Elephant

Materials: The Elephant Scientist

PREP: Read Teacher Tip

→ INTRO
Recall what happened in the last passage.

→ READ, NARRATE, & DISCUSS
p.11-16 "African elephants reign" - "consume enough food."

→ SUPPLEMENTAL
∞ Video Link: Learning to be an Elephant (1:51)

→ NOTE
Read "Elephant Species" on p.17, if desired.

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point: to help generate interest during the introduction, to enliven the lesson itself, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 3

30m Labs: Grade 4 - Lesson 27

Lab: Sending Sound

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 1 of Sending Sound, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in their lab notebook. Any remaining time can be spent collecting the materials and possibly beginning the Procedure up to step 3.

WEEK 4

15m Natural History: Grade 4 - Lesson 28

Surtsey Encounters Life

Materials: Life on Surtsey

→ INTRO
Recall what you read in the last passage.

→ READ, NARRATE, & DISCUSS
p.13-17 "The first visitation" - "that never cools."

→ NOTE
Read "A Lupine is a Lupine" p.14 for interest, if desired.
∞ Video Link: Carl Linnaeus's Systema Naturae (3:33)

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Science: Grade 4

Click THIS text or scan the QR code for links.

Term 3

WEEK 4

⬚ **20m General Science: Grade 4 - Lesson 28**
To Understand an Elephant

Materials: The Elephant Scientist

PREP: Read Teacher Tip

→ INTRO
Recall what happened in the last passage.

→ READ & NARRATE
p.18-19 "Elephants may be" - "in the field."

→ READ, NARRATE, & DISCUSS
p.20-23 "Caitlin and Tim" - "from the area."

• Do you think different parts of your brain help you do different things, like the elephants? Find the temporal lobe in the diagram at the link. Recall what the temporal lobe did for the elephant. Where is your temporal lobe? (Point to it.) Try humming different music that you know. Try a happy song, a sad song, a scary song. Your temporal lobe helps you think about and remember those tunes and how they make you feel!
∞ Image Link: Mayo Clinic: Brain Lobes

★ TEACHER TIP
Again, this reading is presented in smaller portions because this strategy is helpful to some students. Others may be able to read the entire passage before narrating. Do whatever is most appropriate for the student(s).

• FIELD TRIP
Visit a science museum that has a sound exhibit.

WEEK 4

⬚ **30m Labs: Grade 4 - Lesson 28**
Lab: Sending Sound

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Sending Sound, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure today as directed in the Lab Book. Then they will continue to the end, completing the short Analysis and Conclusion with a brief narration.

WEEK 5

⬚ **15m Natural History: Grade 4 - Lesson 29**
Studying Life

Materials: Life on Surtsey

PREP: Read Teacher Tip

→ INTRO
Recall what you read in the last passage. Look at the picture on p.18. Can you imagine flying to this island on a helicopter? That's where our team of scientists is going to study life.

→ NOTE
Read the beginning of this chapter for interest, as desired and as time permits.

→ READ, NARRATE, & DISCUSS
p.23-27 "When scientists" - "of the island."
• Why do you think the arrival of the birds changed this part of the island so much?

★ TEACHER TIP
Are your learners demonstrating familiarity with the natural world as a system where the plants, animals, and nonliving Things all work together? More interest in volcanos, plants, insects, watching how nature changes? If not, consider observing and discussing together, exploring extra helpings, or taking a field trip to a natural history museum.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5

⬚ **20m General Science: Grade 4 - Lesson 29**
Elephant Families

Materials: The Elephant Scientist

PREP: Read Teacher Tip

→ INTRO
Recall what happened in the last passage.

→ READ, NARRATE, & DISCUSS
p.24-28 "After the hectic" - "their unruly behaviors."

• Compare elephant families to human families. How are they similar, and how are they different?

→ SUPPLEMENTAL
∞ Video Link: Elephant Families (2:20)
∞ Video Link: Etosha National Park (3:42)

★ TEACHER TIP
Are your learner(s) demonstrating more ease in thinking about waves or sound? More interest/ease in elephants or animal communication? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a zoo or museum with a sound exhibit. Reach out through Contact Us if you need help.

WEEK 5

⬚ **30m Labs: Grade 4 - Lesson 29**
Lab: The Sound of Your Voice

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 1 of The Sound of Your Voice, as directed in the Lab Book.

Science: Grade 4

Click THIS text or scan the QR code for links.

Term 3

Suggested day 1: Students read the Introduction, compose the prelab narration, and complete step 1 of the Procedure, as directed in the Lab Book.

WEEK 6

⬚ 15m Natural History: Grade 4 - Lesson 30

A Changing Island

Materials: Life on Surtsey

→ INTRO
Recall what you read in the last passage.

→ READ, NARRATE, & DISCUSS
p.27-31 "Some of the" - "on Surtsey."
• Tell about or draw the meaning of the word 'succession' in natural history.
• Can you explain how each change prepared the island for the next change?

→ NOTE
Read "The 2015 Surtsey Team" p.32 for interest, if desired.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6

⬚ 20m General Science: Grade 4 - Lesson 30

The Hypothesis

Materials: The Elephant Scientist

→ INTRO
Recall what happened in the last passage.

→ READ & NARRATE
p.29-33 "It was during" - "alarm call playback."

→ READ, NARRATE, & DISCUSS
p.34-35 "When the dry" - "high-frequency components."

WEEK 6

⬚ 30m Labs: Grade 4 - Lesson 30

Lab: The Sound of Your Voice

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of The Sound of Your Voice, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure today, using the grid lines in their notebook to create a table for data collection. Offer support as needed.

WEEK 7

⬚ 15m Natural History: Grade 4 - Lesson 31

The Spit

Materials: Life on Surtsey

PREP: Read Teacher Tip

→ INTRO
Recall what you read in the last passage. Look at the pictures on p.34, 36 as we walk down to the spit (picture on p.37).

→ NOTE
Read the beginning of this chapter for interest, if desired and as time permits.

→ READ, NARRATE, & DISCUSS
p.37-41 "The spit" - "working with him."
p.44-45 "After a long" - "cooking the dinner!"

→ SUPPLEMENTAL
∞ Video Link: Simple Way to Tell Insects Apart (4:06)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

★ TEACHER NOTE
The supplemental video mentions evolution at 0:53, 1:02, and 1:06. Teachers might choose to allow it to pass by the way, consider if they would like to discuss alternative ideas with their students, or skip this one.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

⬚ 20m General Science: Grade 4 - Lesson 31

Settling into Camp

Materials: The Elephant Scientist

→ INTRO
Recall what happened in the last passage.

Science: Grade 4

Click THIS text or scan the QR code for links.

Term 3

→ READ, NARRATE, & DISCUSS
p.36-41 "As in earlier" - "for the night."

WEEK 7

⬚ 30m Labs: Grade 4 - Lesson 31
Lab: The Sound of Your Voice

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 3 of The Sound of Your Voice, as directed in the Lab Book.

Suggested day 3: Students finish the lab today using the Analysis and Conclusions section. Offer support as needed to construct a bar graph using the grid lines in their notebook. Then complete the postlab narration, as directed.

WEEK 8

⬚ 15m Natural History: Grade 4 - Lesson 32
The Lighthouse

Materials: Life on Surtsey

→ INTRO
Recall what you read in the last passage. Look at the picture on p.46. Can you find the lighthouse? That's where we're going today.

→ READ, NARRATE, & DISCUSS
p.47-52 "In the wee" - "for a picture.'"

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

⬚ 20m General Science: Grade 4 - Lesson 32
The First Experiment

Materials: The Elephant Scientist

→ INTRO
Recall what happened in the last passage.

→ READ, NARRATE, & DISCUSS
p.42-44 "For the first" - "about impending danger."

WEEK 8

⬚ 30m Labs: Grade 4 - Lesson 32
Lab: Talking Animals

Materials: Grade 4 Lab Book and materials listed within

→ LAB DAY
Complete day 1 of Talking Animals, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and compose the prelab narration. Then they review the Procedure and decide when to begin based on the animal's needs.

WEEK 9

⬚ 15m Natural History: Grade 4 - Lesson 33
The Gull Colony

Materials: Life on Surtsey

→ INTRO
Recall what you read in the last passage.

→ READ, NARRATE, & DISCUSS
p.52-55 "The next stop" - "to a close."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9

⬚ 20m General Science: Grade 4 - Lesson 33
More to Learn

Materials: The Elephant Scientist, optional: metal coathanger, metal spoon, 2 pieces of string ~30cm

→ INTRO
Recall what happened in the last passage.

→ READ & NARRATE
p.46-48 "Once Caitlin established" - "no olfactory cue."

→ READ, NARRATE, & DISCUSS
p.49-51 "As Caitlin continued" - "outright killing them."

- Act out the experiment Caitlin did to test how well Donna could sense vibrations.
- Try the demonstration suggested on p.51: “The easiest way to experience this ...” You can have

★ TEACHER NOTE
There is a photograph from the dissection of an elephant's foot on p.50. Most students are not bothered when encountered by the way, and some will find it fascinating. However, if teachers are concerned, they might choose to cover the photo or give students the option to look or not.

Science: Grade 4

[Click THIS text or scan the QR code for links.](#)

Term 3

fun with this by doing the same thing with a metal hanger as shown in the link. Replace the hanger with a metal spoon and see if the sound changes.
∞ Image Link: Coathanger Gong

WEEK 9

⬚ **30m Labs: Grade 4 - Lesson 33**
Lab: Talking Animals

▯ Materials: Grade 4 Lab Book and materials listed within

➔ LAB DAY
Complete day 2 of Talking Animals, as directed in the Lab Book.

Suggested day 2: Students continue the Procedure today, as directed in the Lab Book.

WEEK 10

⬚ **15m Natural History: Grade 4 - Lesson 34**
The Last Day

▯ Materials: Life on Surtsey

➔ INTRO
Recall what you read in the last passage.

➔ READ, NARRATE, & DISCUSS
p.57-62 "The last full" - "every summer."

➔ NOTE
Read "Surtsey's Lava" p.62 for interest, if desired.
∞ Video Link: The Lava Affair (7:53)

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10

⬚ **20m General Science: Grade 4 - Lesson 34**
More Questions

▯ Materials: The Elephant Scientist

➔ INTRO
Recall what happened in the last passage.

➔ READ, NARRATE, & DISCUSS
p.52-59 "A long line" - "his left ear."

• Caitlin's work got her interested in questions about the bulls. How many times can you think of when Caitlin's work created new questions for her to pursue?
• Make up a poem about elephants together!

WEEK 10

⬚ **30m Labs: Grade 4 - Lesson 34**
Lab: Talking Animals

▯ Materials: Grade 4 Lab Book and materials listed within

➔ LAB DAY
Complete day 3 of Talking Animals, as directed in the Lab Book.

Suggested day 3: Students who need to complete the Procedure today may do so and then finish with the Analysis and Conclusions section.

WEEK 11

⬚ **15m Natural History: Grade 4 - Lesson 35**
Departure

▯ Materials: Life on Surtsey

PREP: Read Teacher Tip

➔ INTRO
Recall what you read in the last passage.

➔ READ, NARRATE, & DISCUSS
p.62-65 "He collects" - "this upstart island."
• Explain how new lands sometimes appear in the ocean and then disappear again.

➔ NOTE
Read Appendices p.66-68 for interest, if desired.

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether the ecosystem or any of the new creatures they have met are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Science: Grade 4

[Click THIS text or scan the QR code for links.]

Term 3

WEEK 11

⬚ 20m General Science: Grade 4 - Lesson 35
More Work

Materials: The Elephant Scientist

→ INTRO
Recall what happened in the last passage.

→ READ & NARRATE
p.60-61 "As the sun" - "a helping hand."

→ READ, NARRATE, & DISCUSS
p.62-64 "Elephants once roamed" - "well they match."

→ SUPPLEMENTAL
∞ Video Link: How Elephants Listen (4:33)

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly, whether sound, elephants, or animal communication have become more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us if you need help.

• FIELD TRIP
Visit a zoo that cares for elephants. Read or review Elephant Species p.17 and notice which species is at your zoo.

WEEK 11

⬚ 30m Labs: Grade 4 - Lesson 35
Lab: Catch Up

Materials: Grade 4 Lab Book and materials listed within

→ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

WEEK 12

⬚ 15m Natural History: Grade 4 - Lesson 36
→ EXAMS

Answer question(s) related to course.

Questions will come from:
Life on Surtsey

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 12

⬚ 20m General Science: Grade 4 - Lesson 36
Exam Week

→ EXAMS
Answer question(s) related to course.

Questions will come from:
The Elephant Scientist

WEEK 12

⬚ 30m Labs: Grade 4 - Lesson 36
Lab Exam Week

→ EXAMS
Answer question(s) related to the course.

Questions will come from:
Grade 4 Lab Book

Science: Grade 4

Examination

Term 1

General Science: Grade 4

· The Universe in You: Tell about or draw what microscopes do. OR Tell about or draw something that you or another scientist has seen with microscopes.

· Share what you learned about using the microscope. Tell about a challenge in using this tool and how you worked to overcome this challenge.

Natural History: Grade 4

· A World in a Drop of Water: Describe or draw 3 different examples of microscopic pond life.

Term 2

General Science: Grade 4

· To Fly: Tell about or draw how the Wright Brothers made their plane fly when no one else could. OR Tell how the Wright Brothers helped each other when they got frustrated.

· Explain how you did an investigation in the science lab this term and what you learned from it.

Natural History: Grade 4

· The Octopus Scientists: Tell about or draw 3 unique features of the octopus. OR Tell about why this kind of field work is challenging.

Term 3

General Science: Grade 4

· The Elephant Scientist: What are sound waves? OR Describe what you know about elephant communication.

· Explain how you did an investigation in the science lab this term and what you learned from it.

Natural History: Grade 4

· Life on Surtsey: How are the physical conditions, plant life, and animal life interrelated? OR Explain why it is so important for scientists to study Surtsey.